

Weekly Progress Report
Amity School of Engineering & Technology
B. Tech: Program

WPR No: 5	Duration: 17 August 2026 – 23 August 2026
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Weekly Progress Report (WPR) for Minor Project

To be filled by Students	
Students Name	Rohit Yadav, Vishal, Hardik Mehlawat (Group 148)
Roll no.	472 (Rohit), 568 (Vishal), 449 (Hardik)
Enrollment no.	A2305223472 (Rohit), A2305223568 (Vishal), A2305223449 (Hardik)
Project Title finalized, if Yes, give name, if NO, give reason	Yes — Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches
Synopsis submitted	Submitted (on Amizone, with offer letter)
Literature review	Extraction complete for all 48 primary studies; taxonomy of detection approaches finalised; all 74 reference identifiers verified against arXiv/Crossref records and BibTeX file built
Technical & Economical Feasibility	Not started
Bill of Material	N/A (literature review project this semester)
Project Progress Schedule (PERT Chart)	Week-wise schedule prepared; PERT chart pending
Design of critical components	Paper structure fixed (8 sections); Section 3 taxonomy finalised across subsections 3.1–3.5
Fabrication work (give %)	N/A
Experimental work (give %)	Not started (planned for Semester 8)
Result and Analysis	Not started
Report writing	In progress — Sections 1 (Introduction, 939 words) and 2 (Background, 1,024 words) drafted; Section 3 next
Signature of students	
Work done in this week	
Section 2 (Background) of the review paper was drafted in full this week, at 1,024 words citing 16 verified references. It explains how a generative adversarial network trains a generator and a discriminator in opposition, how diffusion models generate an image through a fixed forward noising process and a learned reverse denoising process, and how latent diffusion moves that process into the compressed latent space of an autoencoder and admits text conditioning through cross-attention. The second half traces where detectable artefacts come from: the traces repeated upsampling leaves in the frequency spectrum, fingerprints specific to an individual model rather than a generator family, the signature of the fixed decoder shared across a latent diffusion system's output, and the fragility of all of these under JPEG compression and ordinary post-processing. The section closes by connecting artefact origin to the cross-generator generalisation problem introduced in Section 1, since evidence that follows from one generator's upsampling operator or decoder is removed, not merely made harder to find, when a new generator changes either. The reference infrastructure was also completed: a BibTeX file covering all 74 included references, keyed by the study identifiers already used in the section drafts. Each of the 74 identifiers was then opened and checked against its source record — arXiv metadata records, Crossref DOI records and one PMLR proceedings page — with one line recorded per reference in the verification log; all 74 are now verified and none pending. One publication-year error was corrected in the process: the LASTED paper had been recorded as 2025, but its DOI resolves to a journal version dated June 2026, and the reference files were updated accordingly. Sections 3.1 and 3.2 are scheduled next.	
To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory ☐	
Performance of students is unsatisfactory ☐	
A warning to be issued to student(s) (Name) _____	
Student was not well (Name) _____	
Date: 26 August 2026	Signature of Guide

Date: 26 August 2026